Chapter 10: A Modern, Dynamic IS-LM Model

Gauti B. Eggertsson, Brown University, June 2, 2020

1 Overview

Here we show a micro-founded version of the IS-LM model. Firms set their prices one period in advance. We assume a particular kind of policy regime, so that in equilibrium all prices are fixed.

We also show that even if a firm could readjust its price, it would choose not to do so if there is even a relatively small cost of changing the price. This remains true even if there are large shocks. In a numerical example, even if there is a 10 percent contraction on output, an individual firm gains only a 0.2 percent increase its profits by changing its price, provided no other firm changes its price. Meanwhile, if all firms change their prices in reaction to the shock, profits would increase by about 6 percent. Thus prices being fixed can be interpreted as a coordination failure, an idea with a long history in macroeconomics.

This variation of the IS-LM model introduces you to the tools needed to consider richer models, in particular models of monopolistic competition.

A key assumption maintained, relative to the standard neoclassical growth model, is that we allow firms to set a price for their product that is different from the price charged by other firms. To model price setting we need to give firms market power. The main technological innovation that allowed economists to study price setting from micro-foundations where firms have market power was a model proposed by Avinash Dixit and Joseph Stiglitz in 1977 [1]. The market structure they proposed allows a tractable way of thinking about the price setting of firms. Below, we assume a utility function that implies the following demand for good i:

$$y_t(i) = \left(\frac{p_t(i)}{P_t}\right)^{-\theta} Y_t,$$

where i indexes a particular good i, and we will be assuming that each good is produced by a single firm. This basic formulation formalizes how a firm can charge a price different from its competitors' price without facing large fluctuations in demand. Recall that in the standard neoclassical growth model firms

take prices as given, while here they make their own optimal pricing decision, resulting in equilibrium markups, i.e. in equilibrium the firm will charge a price above the marginal cost of production. The market structure proposed by Dixit and Stiglitz has a large number of applications in economics (e.g. trade, industrial organization, labor economics, growth theory, etc.). Accordingly, it is well worth your effort to invest considerable time understanding it, much beyond these notes.

2 A Basic Model with Flexible Prices

The household maximizes:

$$\max_{C_t, M_t, B_t} E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma^{-1}}}{1-\sigma^{-1}} + \chi \frac{\left(\frac{M_t}{P_t}\right)^{1-\nu}}{1-\nu} - \frac{{N_t}^{1+\varphi}}{1+\varphi} \right] \xi_t$$

s.t.

$$\int_0^1 p_t(i) c_t(i) di + M_t + B_t = (1+i_{t-1}) B_{t-1} + M_{t-1} + W_t N_t - T_t + \int_0^1 Z_t(i) di$$

Here $c_t(i)$ in the budget constraint refers to consumption of a single good, indexed by i, which the household buys at price $p_t(i)$. The aggregate consumption index C_t that appears in the utility function refers to the composite:

$$C_t \equiv \left[\int_0^1 c_t(i)^{\frac{\theta-1}{\theta}} di \right]^{\frac{\theta}{\theta-1}} \tag{1}$$

where $\theta > 1$. Each good i is produced by a single firm i which has profits $Z_t(i)$. Other notation is similar to what we have seen: M_t is the number of dollars the household holds, W_t is the nominal wage rate, N_t is labor supply, ξ_t is a preference shock, T_t is lump sum taxes, B_t is a risk-free nominal bond that pays i_t number of dollars in next period. Note that we suppose that the household owns the firms.

2.1 The Household Problem

The easiest way to solve the household problem is in two steps:

1. **Static Problem**: derive the choice of the agents across the differentiated goods given a certain budget at a certain point in time;

2. **Dynamic Problem**: given the solution to 1, solve the dynamic optimization problem, i.e. obtain the intertemporal choice of consumption

across time. As it will turn out – and this is the key advantage of the Dixit-Stiglitz model – we end up with exactly the same problem as before, except that now the interpretation of C_t, etc. is that they refer to aggregate quantities.

Step 1. We first find the optimal spending allocation by solving:

$$\begin{aligned} \max\ C_t &= \left[\int_0^1 c_t(i)^{\frac{\theta-1}{\theta}} di\right]^{\frac{\theta}{\theta-1}} \\ s.t. \quad E &= \int_0^1 p_t(i)c_t(i)di \end{aligned}$$

We can solve the household problem setting up the Lagrangian:

$$\mathcal{L}_0 = \left[\int_0^1 c_t(i)^{\frac{\theta-1}{\theta}} di\right]^{\frac{\theta}{\theta-1}} - \lambda\left[E - \int_0^1 p_t(i)c_t(i)di\right]$$

First-order conditions (FOC) are obtained by taking the partial derivative of the Lagrangian subject to each of the decision variables:

$$\begin{aligned} \frac{\partial \mathcal{L}_0}{\partial c_t(i)} &= \frac{\theta}{\theta-1}\left[\int_0^1 c_t(i)^{\frac{\theta-1}{\theta}} di\right]^{\frac{\theta}{\theta-1}-1} \frac{\theta-1}{\theta} c_t(i)^{\frac{\theta-1}{\theta}-1} + \lambda p_t(i) \\ &= \left[\int_0^1 c_t(i)^{\frac{\theta-1}{\theta}} di\right]^{\frac{1}{\theta-1}} c_t(i)^{-\frac{1}{\theta}} + \lambda p_t(i) = 0 \end{aligned}$$

If we take the ratio of the above FOC for two goods, say i and j, we get:

$$\begin{aligned} \frac{\left[\int_0^1 c_t(i)^{\frac{\theta-1}{\theta}} di\right]^{\frac{1}{\theta-1}} c_t(i)^{-\frac{1}{\theta}}}{\left[\int_0^1 c_t(j)^{\frac{\theta-1}{\theta}} dj\right]^{\frac{1}{\theta-1}} c_t(j)^{-\frac{1}{\theta}}} &= \frac{-\lambda p_t(i)}{-\lambda p_t(j)} \\ \left(\frac{c_t(i)}{c_t(j)}\right)^{-\frac{1}{\theta}} &= \frac{p_t(i)}{p_t(j)} \end{aligned}$$

and finally:

$$c_t(i) = \left(\frac{p_t(i)}{p_t(j)}\right)^{-\theta} c_t(j)$$

To write the demand for the single consumption good in terms of aggregate consumption, use the definition of the consumption index $C_t \equiv \left[\int_0^1 c_t(i)^{\frac{\theta-1}{\theta}} di\right]^{\frac{\theta}{\theta-1}}$

and plug in the optimal demand just derived:

$$\begin{aligned} C_t &\equiv \left[\int_0^1 \left(\left(\frac{p_t(i)}{p_t(j)}\right)^{-\theta} c_t(j)\right)^{\frac{\theta-1}{\theta}} di\right]^{\frac{\theta}{\theta-1}} \\ &= \left[\int_0^1 \left(\frac{p_t(i)}{p_t(j)}\right)^{1-\theta} c_t(j)^{\frac{\theta-1}{\theta}} di\right]^{\frac{\theta}{\theta-1}} \\ &= \left[\int_0^1 p_t(i)^{1-\theta} di\right]^{\frac{\theta}{\theta-1}} \left(\frac{1}{p_t(j)}\right)^{-\theta} c_t(j) \end{aligned}$$

Defining the price aggregate as:

$$P_t \equiv \left[\int_0^1 p_t(i)^{1-\theta} di\right]^{\frac{1}{1-\theta}} \tag{2}$$

we get $C_t = P_t^{-\theta} p_t(j)^{\theta} c_t(j)$ and hence:

$$c_t(i) = \left(\frac{p_t(i)}{P_t}\right)^{-\theta} C_t$$

which is the **demand for good i**. Note from this expression how θ can be interpreted as the price elasticity of demand.

Step 2. To solve the problem, we will rewrite the budget constraint in terms of the aggregate variables. Compute total expenditure:

$$\begin{aligned} \int_0^1 p_t(i) c_t(i) di &= \int_0^1 p_t(i) \left(\frac{p_t(i)}{P_t}\right)^{-\theta} C_t di \\ &= C_t P_t^{\theta} \int_0^1 \left(p_t(i)\right)^{1-\theta} di \\ &= C_t P_t^{\theta} P_t^{1-\theta} = P_t C_t \end{aligned}$$

Hence, we can rewrite the budget constraint as:

$$P_t C_t + M_t + B_t = (1 + i_{t-1}) B_{t-1} + M_{t-1} + W_t N_t - T_t + \int_0^1 Z_t(i) di$$

To obtain the FOCs of the representative household, form a Lagrangian:

$$\begin{aligned} \mathcal{L}_0 = & E_0 \sum_{t=0}^{\infty} \beta^t \Bigg\{ \left[\frac{C_t^{1-\sigma^{-1}}}{1-\sigma^{-1}} + \chi \frac{\left(\frac{M_t}{P_t}\right)^{1-\nu}}{1-\nu} - \frac{N_t^{1+\varphi}}{1+\varphi}\right] \xi_t + \phi_t (P_t C_t + M_t + B_t \\ & - (1 + i_{t-1}) B_{t-1} - M_{t-1} - W_t N_t + T_t - \int_0^1 Z_t(i) di) \Bigg\} \end{aligned}$$

The FOCs of this Lagrangian are:

$$\frac{\partial \mathcal{L}_0}{\partial C_t} = \beta^t C_t^{-\sigma^{-1}} \xi_t + \beta^t \phi_t P_t = 0$$
$$\frac{\partial \mathcal{L}_0}{\partial B_t} = \beta^t \phi_t - \beta^{t+1}(1+i_t) E_t \phi_{t+1} = 0$$
$$\frac{\partial \mathcal{L}_0}{\partial M_t} = \beta^t \chi \left(\frac{M_t}{P_t}\right)^{-\nu} \frac{1}{P_t} \xi_t + \beta^t \phi_t - \beta^{t+1} E_t \phi_{t+1} = 0$$
$$\frac{\partial \mathcal{L}_0}{\partial N_t} = -\beta^t N_t{}^{\varphi} \xi_t - \beta^t \phi_t W_t = 0$$

These can be combined to yield:

i. **Consumption Euler Equation:**

$$\frac{1}{1+i_t} = \beta E_t \frac{C_{t+1}^{-\sigma^{-1}}}{C_t^{-\sigma^{-1}}} \frac{\xi_{t+1}}{\xi_t} \frac{P_t}{P_{t+1}}$$

ii. **Money Demand:**

$$\chi \frac{\frac{M_t}{P_t}^{-\nu}}{C_t^{-\sigma^{-1}}} = \frac{i_t}{1+i_t}$$

iii. **Labor Supply:**

$$\frac{W_t}{P_t} = \frac{N_t^{\varphi}}{C_t^{-\sigma^{-1}}}$$

2.2 Firm's Optimization Problem

Firms can now set a price for the good they sell. The firm's problem is to maximize expected profits given by:

$$E_t \sum_{s=t}^{\infty} Q_{t,s} Z_s(i),$$

where $Q_{t,t+s} = \beta^s \frac{C_{t+s}^{-\sigma^{-1}}}{C_t^{-\sigma^{-1}}} \frac{\xi_{t+s}}{\xi_t} \frac{P_t}{P_{t+s}}$ is the stochastic discount factor that the household uses at time t to valuate nominal income received in time $t+s$, as discussed in Chapter 4. At time period t, profits are given by:

$$Z_t(i) = p_t(i) y_t(i) - W_t N_t(i)$$

The firm maximizes profits subject to technology: $y_t(i) = A_t N_t(i)^{1-\alpha}$, where A_t is the level of technology that evolves stochastically. The firm also faces the demand: $y_t(i) = \left(\frac{p_t(i)}{P_t}\right)^{-\theta} Y_t$, where consumption has been replaced by output because we are assuming that all the production is consumed.

We can rewrite the profits in terms of the price of the firm's good, so that the maximization will be with respect to only one variable, $p_t(i)$, a simplification that is useful for more complex problems. Substitute for $N_t(i) = \left(\frac{y_t(i)}{A_t}\right)^{\frac{1}{1-\alpha}}$ and then express $y_t(i)$ in terms of the price using the demand function, to express the profit in terms of the price:

$$\begin{aligned} Z_t(i) &= p_t(i)y_t(i) - W_t\left(\frac{y_t(i)}{A_t}\right)^{\frac{1}{1-\alpha}} \\ &= p_t(i)^{1-\theta}P_t^{\theta}Y_t - W_t p_t(i)^{-\frac{\theta}{1-\alpha}}\left(\frac{P_t^{\theta}Y_t}{A_t}\right)^{\frac{1}{1-\alpha}} \end{aligned} \tag{3}$$

Each period profit does not depend on any variable at time t or $t-1$. Hence, if prices are fully flexible, the problem of the firm is a sequence of static maximization problems for each period t, which can be written as:

$$\max_{p_t(i)} \; Z_t(i) = p_t(i)^{1-\theta}P_t^{\theta}Y_t - W_t p_t(i)^{-\frac{\theta}{1-\alpha}}\left(\frac{P_t^{\theta}Y_t}{A_t}\right)^{\frac{1}{1-\alpha}}$$

from which we get the FOC:

$$\frac{\partial Z_t(i)}{\partial p_t(i)} = (1-\theta)p_t(i)^{-\theta}P_t^{\theta}Y_t - \frac{\theta}{1-\alpha}W_t p_t(i)^{-\frac{\theta}{1-\alpha}-1}\left(\frac{P_t^{\theta}Y_t}{A_t}\right)^{\frac{1}{1-\alpha}} = 0$$

This can be rewritten (with some manipulation) to yield:

$$\frac{p_t(i)}{P_t} = \frac{\theta}{\theta-1}\frac{W_t}{P_t}\frac{1}{1-\alpha}N_t(i)^{\alpha}\frac{1}{A_t} \tag{4}$$

which has a useful interpretation. The term $\frac{\theta}{\theta-1}$ is the *markup* of the firm, i.e. it tells us how much the firm will charge for the good it is producing in excess of its marginal cost. The marginal cost of production is given by the remainder of the right-hand side and is decomposed as follows. $\frac{W_t}{P_t}$ is the wage bill for hours needed to produce one more unit of output. The term $\frac{1}{1-\alpha}N_t(i)^{\alpha}\frac{1}{A_t}$, however, represents how many hours of labor input are needed to produce one more unit of output of the firm. Accordingly, $\frac{W_t}{P_t}\frac{1}{1-\alpha}N_t(i)^{\alpha}\frac{1}{A_t}$ represents the firm's marginal cost of production. Observe that, if $\alpha = 0$ and $A_t = 1$, then the production function is linear in labor and one "hour" (or whatever unit we use to define the labor input) of labor produces one unit of output. In this case, the marginal cost is simply the wage rate $\frac{W_t}{P_t}$.

Under the assumption that all prices are flexible, each firm faces the exact same problem and will accordingly set the same price, so that $p_t(i) = P_t$, $y_t(i) = Y_t$ and $N_t(i) = N_t$ for every i and we obtain a labor demand equation:

$$\frac{W_t}{P_t} = \frac{\theta-1}{\theta}(1-\alpha)A_t^{\frac{1}{1-\alpha}}Y_t^{-\frac{\alpha}{1-\alpha}}$$

2.3 Equilibrium under Flexible Prices: A Classical Dichotomy

If fiscal policy is sustainable in the way we defined before, we can prove Ricardian Equivalence and that the choice of taxes is irrelevant. The model is summarized in the following definition of equilibrium, where once again we have defined $m_t \equiv \frac{M_t}{P_t}$, $\mu_t \equiv \frac{M_t}{M_{t-1}}$, $\Pi_t \equiv \frac{P_t}{P_{t-1}}$.

Definition 1. *An equilibrium under sustainable fiscal policy with flexible prices is a collection of stochastic processes for* $\{m_t, \frac{W_t}{P_t}, C_t, \Pi_t, i_t, Y_t, \mu_t\}$ *for* $t \geq 0$ *that satisfy (5)-(10), given an exogenous path for* $\{A_t, \xi_t\}$ *and a specification for* $G(.)$ *for fiscal policy and* $i(.)$ *or* $\mu(.)$ *for monetary policy, given initial condition* m_{-1}.

The equilibrium equations are:

Consumption Euler:

$$\frac{1}{1+i_t} = \beta E_t \frac{C_{t+1}^{-\sigma^{-1}}}{C_t^{-\sigma^{-1}}} \frac{\xi_{t+1}}{\xi_t} \Pi_{t+1}^{-1} \tag{5}$$

Money Demand:

$$\chi \frac{m_t^{-\nu}}{C_t^{-\sigma^{-1}}} = \frac{i_t}{1+i_t} \tag{6}$$

Money Identity:

$$m_t = m_{t-1} \mu_t \Pi_t^{-1} \tag{7}$$

Labor Supply:

$$\frac{W_t}{P_t} = \frac{Y_t^{\frac{\varphi}{1-\alpha}} A_t^{-\frac{\varphi}{1-\alpha}}}{(Y_t - G_t)^{-\sigma^{-1}}} \tag{8}$$

Labor Demand:

$$\frac{W_t}{P_t} = \frac{\theta - 1}{\theta} (1-\alpha) A_t^{\frac{1}{1-\alpha}} Y_t^{-\frac{\alpha}{1-\alpha}} \tag{9}$$

Resource Constraint:

$$Y_t = C_t + G_t \tag{10}$$

The following proposition immediately follows:

Proposition 1 (Classical Dichotomy Between Output and Inflation)**.** *Output is determined independently of monetary policy.*

This can be proved by combining (8), the labor supply, and (9), the labor demand, which then will implicitly define output as a solution to the following equation:

$$(Y_t - G_t)^{\sigma^{-1}} Y_t^{\frac{\varphi+\alpha}{1-\alpha}+} = \frac{\theta-1}{\theta}(1-\alpha)A_t^{\frac{1+\varphi}{1-\alpha}} \tag{11}$$

These equations that define the equilibrium in the non-linear model can be approximated to define an approximate solution to the model. Note that we can combine the labor demand and supply equations as in (11). We obtain:

$$\hat{Y}_t - \hat{G}_t = E_t\hat{Y}_{t+1} - E_t\hat{G}_{t+1} - \sigma_c(\hat{\imath}_t - E_t\pi_{t+1} - \hat{\xi}_t + E_t\hat{\xi}_{t+1}) \tag{12}$$

$$\hat{Y}_t = \frac{1+\varphi_\alpha}{\sigma_c^{-1}+\varphi_\alpha}\hat{A}_t + \frac{\sigma_c^{-1}}{\sigma_c^{-1}+\varphi_\alpha}\hat{G}_t \tag{13}$$

$$\hat{m}_t = \eta_y\hat{Y}_t - \eta_i\hat{\imath}_t \tag{14}$$

$$\hat{m}_t = \hat{m}_{t-1} + \hat{\mu}_t - \pi_t \tag{15}$$

where $\sigma_c = \sigma\frac{\bar{C}}{\bar{Y}}$ and $\varphi_\alpha = \frac{\alpha+\varphi}{1-\alpha}$. All variables are defined as $\hat{x}_t \equiv \frac{x_t-\bar{x}}{\bar{x}}$, except for government spending, which is defined as a ratio of steady state output, i.e. $\hat{G}_t \equiv \frac{G_t-\bar{G}}{\bar{Y}}$.

We assume that government spending is distributed across the differentiated goods in exactly the same way as private spending. Similarly, for simplicity, in the approximated equation (14) we assume a symmetric money demand function for the government as for the private agents, so that (14) is a function of aggregate output rather than private consumption.[1] In coming chapters we will consider variations on the basic assumption of price setting, which will result in a different characterization of the labor demand by the firms given by (9). This, together with specific assumptions about monetary and fiscal policies, leads to alternative models.

3 A Microfounded IS-LM Model

We first show how the model, with appropriate assumptions about price setting, can be reduced to the standard IS-LM model, even if there remain some differences. Consider the previously derived model with flexible prices, but now assume that, instead of being flexible, all prices are set one period in advance. As prices are set one period in advance, we need to set the price level taking into account expectations about the future price. The price setting problem of the firm will again boil down to a sequence of static maximization problems,

[1] i.e. we assume: $\chi\frac{(m_t^g)^\nu}{(G_t)^{-\sigma^{-1}}} = \frac{i_t}{1+i_t}$, which implies that the linearized money demand involves output and not consumption.

which are now given by:

$$\max_{p_t(i)} \; E_{t-1} Q_{t-1,t} Z_t(i)$$

Substituting the stochastic discount factor (note that we can bring the terms indexed to $t-1$ outside the expectation and, since they do not depend on $p_t(i)$, they can be left out) and $Z_t(i)$, the problem becomes:

$$\max_{p_t(i)} \; E_{t-1}\beta \frac{{C_t}^{-\sigma^{-1}}\xi_t}{P_t}\left\{ p_t(i)^{1-\theta} P_t^{\theta} Y_t - W_t p_t(i)^{-\frac{\theta}{1-\alpha}} \left(\frac{P_t^{\theta} Y_t}{A_t}\right)^{\frac{1}{1-\alpha}} \right\}$$

yielding the following first-order condition:

$$\begin{aligned}\frac{\partial E_{t-1} Q_{t-1,t} Z_t(i)}{\partial p_t(i)} =& E_{t-1}\beta \frac{C_t^{-\sigma^{-1}}\xi_t}{P_t}\Bigg\{ (1-\theta) p_t(i)^{-\theta} P_t^{\theta} Y_t \\ &+ \frac{\theta}{1-\alpha} W_t p_t(i)^{-\frac{\theta}{1-\alpha}-1} \left(\frac{P_t^{\theta} Y_t}{A_t}\right)^{\frac{1}{1-\alpha}} \Bigg\} = 0 \end{aligned} \tag{16}$$

Since all firms behave in the same way, then $p_t(i) = P_t$, so this condition can be simplified to:

Labor Demand:

$$E_{t-1} C_t^{-\sigma^{-1}} Y_t \xi_t \left\{ 1 - \frac{\theta}{\theta - 1} \frac{1}{1-\alpha} \frac{W_t}{P_t} \left(\frac{Y_t^{\alpha}}{A_t}\right)^{\frac{1}{1-\alpha}} \right\} = 0 \tag{17}$$

The equilibrium is now determined by (5)-(10), but with (9) replaced by (17). The linearized set of conditions replaces (13) with:

$$E_{t-1}(\hat{Y}_t - \hat{Y}_t^n) = 0 \tag{18}$$

$$\hat{Y}_t^n \equiv \frac{1+\varphi_{\alpha}}{\sigma_c^{-1}+\varphi_{\alpha}} \hat{A}_t + \frac{\sigma_c^{-1}}{\sigma_c^{-1}+\varphi_{\alpha}} \hat{G}_t \tag{19}$$

The exogenous term $\hat{Y}_t^n$ has the useful interpretation of corresponding to output if prices are flexible, as we discuss in more detail in the next chapter.

Consider now the following monetary and fiscal policies:

$$\hat{M}_t = \varepsilon_t^m \tag{20}$$

$$\hat{G}_t = \epsilon_t^g \tag{21}$$

where ϵ_t^g and ϵ_t^m are iid mean zero processes. We now have an IS-LM model in the approximated form.

Proposition 2. *If all prices are set one period in advance as in (16), if we assume a stable policy regime of the form (20) and (21) and an iid process for* $\hat{A}_t$ *and* $\hat{\xi}_t$*, then, in the approximated equilibrium,* $\hat{P}_t = 0$ *at all times and the model reduces to the IS-LM model given by the IS equation:*

$$\hat{Y}_t = \hat{G}_t - \sigma_c \hat{\imath}_t + \sigma_c \hat{\xi}_t$$

and the LM equation:

$$\hat{M}_t = \eta_y \hat{Y}_t - \eta_i \hat{\imath}_t$$

The proposition is proved by observing that (18) implies that $E_t \hat{Y}_{t+1} = E_t \hat{Y}^n_{t+1}$. Then, the equations (12) and (14) of the approximated equilibrium can be combined to yield:

$$\begin{aligned}\hat{P}_t = &\left(\frac{\eta_i}{1+\eta_i}\right)^T E_t \hat{P}_{t+T} + E_t \sum_{s=0}^{T-1} \left(\frac{\eta_i}{1+\eta_i}\right)^{s+1} [\sigma_c \hat{Y}^n_{t+s+1} + \sigma_c^{-1}(\hat{G}_{t+s} - \hat{G}_{t+s+1}) \\ &- (\sigma_c^{-1} + \eta_y \eta_i^{-1}) \hat{Y}_{t+s} + \eta_i^{-1} \hat{M}_{t+s} + \hat{\xi}_{t+s} - \hat{\xi}_{t+s+1}]\end{aligned}$$

Hence, there is a unique bounded solution as long as $\eta_i > 0$, which is given by:

$$\begin{aligned}\hat{P}_t = &E_t \sum_{s=0}^{\infty} \left(\frac{\eta_i}{1+\eta_i}\right)^{s+1} [\sigma_c^{-1} \hat{Y}^n_{t+s+1} + \sigma_c^{-1}(\hat{G}_{t+s} - \hat{G}_{t+s+1}) - (\sigma_c^{-1} + \eta_y \eta_i^{-1}) \hat{Y}_{t+s} \\ &+ \eta_i^{-1} \hat{M}_{t+s} + \hat{\xi}_{t+s} - E_t \hat{\xi}_{t+s+1}]\end{aligned}$$

Let us suppose that $\hat{A}_t$ and $\hat{\xi}_t$ are zero mean *iid* processes, in which case $E_t \hat{Y}_{t+1} = 0$. Then, this equation simplifies to:

$$\hat{P}_t = \frac{\eta_i}{1+\eta_i} [\sigma_c^{-1} \hat{G}_t + (\sigma_c^{-1} + \eta_y \eta_i^{-1}) \hat{Y}_t + \eta_i^{-1} \hat{M}_t + \hat{\xi}_t - E_t \hat{\xi}_{t+1}]$$

Take expectation at time $t-1$ from both sides to obtain $E_{t-1} \hat{P}_t = 0$. Since prices are set one period in advance, this implies that they are known at time $t-1$ and, therefore, that $\hat{P}_t = E_{t-1} \hat{P}_t = 0$, i.e. prices are constant at all times, as in the IS-LM model. Using this result, we can rewrite the system as:

$$\begin{aligned}\hat{Y}_t &= \hat{G}_t - \sigma_c \hat{\imath}_t + \sigma_c \hat{\xi}_t \\ \hat{M}_t &= \eta_y \hat{Y}_t - \eta_i \hat{\imath}_t\end{aligned}$$

which proves the proposition. These two equations are plotted in Figure 1. The IS-LM model has several major predictions: an increase in M shifts the LM schedule to the right, thus reducing interest rate and increasing output. Meanwhile, an increase in government spending shifts out the IS curve, thus increasing interest rate and output.

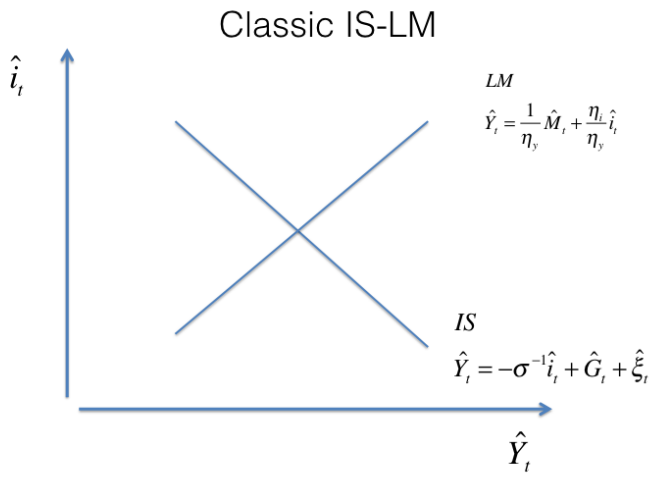


Figure 1: IS-LM Model

There are several important differences with the standard IS-LM model. For example, due to the Ricardian Equivalence, taxes do not appear in the IS equation. Another important implication is that this model is completely dependent on the stability of the policy regime, but we assumed that $\hat{M}_t$ and $\hat{G}_t$ are iid. What happens here if there is an unexpected permanent increase in the money supply? In this case there is only an effect on output today, while output does not respond tomorrow when instead prices increase. More generally, if there is an increase in the money supply that is anticipated, there is no effect on output, only on prices. Hence, the fundamental assumption underlying our characterization of the IS-LM model, derived from microfoundations, is that it applies under a stable regime where any shock is temporary and thus firms will keep their prices unchanged.

3.1 What is the Cost of Keeping Prices Unchanged?

Is it reasonable to assume that firms will maintain their price, even in the face of a large negative shock? The model assumed so, but we can also ask whether it is optimal for firms to maintain their price, conditional on all other firms doing the same, by introducing a small cost of changing the price. Under reasonable assumptions about the parameters of the model, we show below that it can be rational for the firms to maintain their price fixed *even if the cost of changing*

the price is very small, provided that nobody else is changing their price. This may apply even if the social welfare loss of fixed prices is very large, an insight that dates back to Mankiw (1985) [2], and *even if firms collectively could gain considerably by coordinating on a price cut.*

Consider the real profits of firm i at time t. The point is most clearly made by considering a policy regime that gives rise to variations in nominal demand D_t, defined as $D_t = Y_t P_t$. If all prices are fixed at some predetermined level, then an increase in nominal demand D_t leads to an increase in aggregate output, Y_t, one-to-one. Our question is: how much profit does firm i forgo by not changing its price, assuming all other firms do not change their price? Hence, we are considering the effect on profits of $p_t(i)$ set optimally and different from P_t. For simplicity, we assume here that $G_t = 0$ and $A_t = 1$. This implies that, from (8) and (9), the steady state output $\bar{Y}$ and the steady real state wage, defined as $\bar{w} = \frac{\bar{W}}{\bar{P}}$, are given by:

$$\bar{Y} = \left[\frac{\theta - 1}{\theta}(1-\alpha)\right]^{\frac{1}{\sigma^{-1} + \frac{\varphi+\alpha}{1-\alpha}}} \tag{22}$$

$$\bar{w} = \bar{Y}^{\sigma^{-1} + \frac{\varphi}{1-\alpha}} \tag{23}$$

The result is starkest if we assume that the real wage rate is rigid, i.e. that $\frac{W_t}{P_t} = \bar{w}$ in response to changes in D_t. In the general model we have considered, this would only be true under the extreme case where $\sigma^{-1} = \varphi = 0$. More generally, there are several reasons why real wages may only move slowly. Recall that we are exploring an equilibrium in which the prices are constant in response to a negative drop in aggregate demand. Accordingly, one reason why real wages do not change could be that the nominal wage rate cannot adjust downward for some reason – an assumption common in the old Keynesian literature and argued by some to be a stylized fact of the data. Under the assumption that $\frac{W_t}{P_t} = \bar{w}$, profits of firm i are given by:

$$\begin{aligned} z_t(i) = \frac{Z_t(i)}{P_t} &= \frac{p_t(i)}{P_t} y_t(i) - \frac{W_t}{P_t} N_t(i) \\ &= \left(\frac{p_t(i)}{P_t}\right)^{1-\theta} \left(\frac{D_t}{P_t}\right) - \bar{w} \left(\frac{p_t(i)}{P_t}\right)^{-\frac{\theta}{1-\alpha}} \left(\frac{D_t}{P_t}\right)^{\frac{1}{1-\alpha}} \end{aligned}$$

where we have used the definition of demand faced by firm i: $y_t(i) = \left(\frac{p_t(i)}{P_t}\right)^{-\theta} Y_t$; and, from the definition of the production function, again using the demand function: $N_t(i) = y_t(i)^{\frac{1}{1-\alpha}} = \left(\frac{y_t(i)}{Y_t}\right)^{\frac{1}{1-\alpha}} Y_t^{\frac{1}{1-\alpha}} = \left(\frac{p_t(i)}{P_t}\right)^{-\frac{\theta}{1-\alpha}} Y_t^{\frac{1}{1-\alpha}}$. The profits that firm i obtains if it sets its price to the same level as other firms, i.e.

$p_t(i) = P_t$, are:

$$z_t^{fixed}(i) = \left(\frac{D_t}{P_t}\right) - \bar{w}\left(\frac{D_t}{P_t}\right)^{\frac{1}{1-\alpha}} \tag{24}$$

Consider now the profits if firm i changes its price. The optimal price, p_t^*, satisfies equation (4):

$$\begin{aligned}\frac{p_t^*}{P_t} &= \frac{\theta}{\theta-1}\bar{w}\frac{1}{1-\alpha}N_t(i)^{\alpha} \\ &= \frac{\theta}{\theta-1}\bar{w}\frac{1}{1-\alpha}\left(\frac{p_t^*}{P_t}\right)^{-\frac{\theta\alpha}{1-\alpha}} Y_t^{\frac{\alpha}{1-\alpha}}\end{aligned}$$

where we have substituted again for $N_t(i)$. Solving for $\frac{p_t^*}{P_t}$ yields:

$$\frac{p_t^*}{P_t} = \left(\frac{\theta}{\theta-1}\bar{w}\frac{1}{1-\alpha}Y_t^{\frac{\alpha}{1-\alpha}}\right)^{\frac{1-\alpha}{1-\alpha+\theta\alpha}},$$

which we can substitute into the profit function to derive the profits from changing prices:

$$z_t(i)^{adjust} = \left(\frac{p_t^*}{P_t}\right)^{1-\theta}\left(\frac{D_t}{P_t}\right) - \bar{w}\left(\frac{p_t^*}{P_t}\right)^{-\frac{\theta}{1-\alpha}}\left(\frac{D_t}{P_t}\right)^{\frac{1}{1-\alpha}} \tag{25}$$

Consider now the following numerical example: $\sigma^{-1} = 1$, $\varphi = 1$, $\theta = 5$, $\alpha = 0.25$. By equation (22) and (23), these values imply $\bar{Y} = 0.82$ and $\bar{w} = 0.64$. Hence, steady state profits relative to output, $\frac{\bar{z}}{\bar{Y}}$, are 40%. Consider now a 10% reduction in nominal demand D_t. If all firms change their prices flexibly, this will leave output and profits unchanged. Now consider the case in which all prices are fixed. Then the shock generates a 10% reduction in output, Y_t. Using (24), this results in a reduction in the profits of firms – in the aggregate – by 5.4%, a lower number with respect to the reduction in output. The reason is that firms now have a higher markup, given the fixed price, as their marginal costs have gone down due to decreasing returns of production. But shouldn't this give firms strong incentives to change prices, as this would prevent the fall in profits by 5.4%? They key insight of the model is that even if it is in the collective interest of the firms that they all change their prices, each firm does not have a strong incentive to change its price if nobody else is changing theirs. Consider the incentives of firm i to change prices, conditional on nobody else changing their own prices. Using formula (25), this leads to an increase in profits, relative to not adjusting, of only 0.2%! Thus, there is a coordination problem. Even if all firms would gain from everyone changing their prices, the incentives for each one of them to change its price alone are not very strong.

This result depends on real wages not responding to the drop in nominal demand. In the full model, however, the real wage rate does respond to a drop in output if $\sigma^{-1} > 0$ and $\varphi > 0$, as the labor supply equation is given by:

$$\frac{W_t}{P_t} = Y_t^{\sigma^{-1} + \frac{\varphi}{1-\alpha}}$$

A 10% drop in output, according to this equation, with the parameter values given above, would yield a drop in the real wage by 20%. This gives firms much stronger incentives to cut prices – as production costs have gone down – thus increasing firm i's profits for changing prices by 11% with respect to not adjusting (relative to 0.2% with a fixed real wage). This number could be reduced, for example, by lowering the values of φ and σ^{-1} or, of course, by a smaller shock to nominal GDP. Consider, for example, a 3% drop in nominal GDP. Then, the incentives for a firm to change its price correspond to a 1% increase in its profits, relative to not adjusting. Moving to lower values of φ and σ^{-1}, such as 0.5 for both, then profits increase by 4% for a 10% drop in GDP (relative to 11%) and by 0.3% for a 3% reduction in nominal GDP (relative to 1%), as a result of cutting prices, with respect to not adjusting.

Generally, the strength of the incentive for the firm to change its price depends fundamentally on how sensitive its profits are to changing its price, conditional on other firms not changing theirs, something called strategic complementarity in price setting or real rigidities. The key point is that, if the rigidities are strong enough, then the incentive to change prices can be very small for each firm, even if the social loss of the recession is very large and even if it is in their own joint interest to change their prices. This gives a firmer rationale for why keeping prices unchanged, as assumed in the dynamic IS-LM model we just derived, may not be a bad assumption, provided the shocks are not too big.

4 References

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5 Problems

Problem 1
Each firm faces a demand for its product, $y_t(i)$, given by $y_t(i) = \left(\frac{p_t(i)}{P_t}\right)^{-\theta} Y_t$, where $\theta > 1$, Y_t is aggregate output and P_t aggregate price level. Each firm produces using a production function $y_t(i) = h_t(i)^{1-\alpha}$, where $h_t(i)^{1-\alpha}$ is labor it hires from a common labor market. Each firm hires labor at a nominal wage rate W_t per hour. What is the markup a firm charges for each unit of the good it produces, i.e. how much does it charge for the product in excess of what it costs to produce that same unit?

Problem 2
Consider a linear macro model consisting of the following equations:

$$\hat{m}_t - \hat{p}_t = \eta_y \hat{y}_t - \eta_i \hat{\imath}_t \tag{26}$$

$$\hat{y}_t = E_t \hat{y}_{t+1} - \chi[\hat{\imath}_t - E_t(\hat{p}_{t+1} - \hat{p}_t)] + \hat{\delta}_t \tag{27}$$

$$E_t \hat{y}_{t+1} = 0 \tag{28}$$

In these equations, $\hat{m}_t$ represents money supply, $\hat{p}_t$ the price level, $\hat{y}_t$ output, $\hat{\imath}_t$ the nominal interest rate, while $\hat{\delta}_t$ is an iid random disturbance with mean zero; η_y, η_i and χ are coefficients. All variables are expressed in percentage deviation from steady state. This set of equations is derived from a micro-founded model in which all prices are set one period in advance. This assumption implies that:

$$E_t \hat{p}_{t+1} = \hat{p}_{t+1} \tag{29}$$

We assume that money supply is set by the central bank so that:

$$\hat{m}_t = \gamma \hat{m}_{t-1} + \psi \hat{\delta}_{t-1} \tag{30}$$

where $0 < \gamma < 1$ and ψ is a policy parameter.

i. Briefly explain the economic interpretation of equations (26)-(28) in terms of an optimizing model in which prices are fixed one period in advance. Explain the economic significance of parameters η_y, η_i and χ and what sign you would expect them to have (and why).

ii. Show that there is a unique bounded rational expectation solution to the system above. ***Hint:*** 1) First solve the policy rule (30) to obtain an expression for $E_t\hat{m}_{t+j}$ for $j > 0$ as a function of m_t and δ_t. 2) Then, combine (26)-(29) to solve for the unique bounded solution for $\hat{p}_{t+1}$, which is determined at time t by (29), as a function of current and expected future values of $\hat{m}_{t+j}$. To do this, note that you can forward (26) and (27) one period, take conditional expectation E_t, and use (29) to obtain a difference equation in only p_{t+1}, $E_t\hat{p}_{t+2}$ and $E_t\hat{m}_{t+1}$, without any reference to $\hat{y}_t$. Then use your solution from part 1) of the hint to obtain an expression of $\hat{p}_t, \hat{y}_t, \hat{\imath}_t$.

iii. Show how the response of output to the disturbance $\hat{\delta}_t$ depends on the central bank's choice of ψ. Given that the central bank is assumed not to be able to respond to the shock until the following period (it is $\hat{m}_{t+1}$, not $\hat{m}_t$, that responds to $\hat{\delta}_t$) and that prices in the economy are able to adjust by that time, why doesn't rational expectation by the price setters imply that price adjustment eliminates the real effect of monetary policy? What value of ψ creates the smallest variability in output?